Full Project Documentation & Engineering Handbook

Project: Health Insurance Fraud Intelligence Platform (`Medical Insurance Claim Fraud Detection`)

Institution: Department of Computer Science and Engineering, IIIT Dharwad

Authors: B. Varshith (`23BDS011`), J. Ganesh Kumar Reddy (`23BDS024`), M. Jagadeeswar (`23BDS033`)

Faculty Guide: Prof. Ramesh Athe

1. Repository Map & Module Ownership

Directory / File	Approach / Role	Primary Owner	Status & Key Contents
`Health Insurance Fraud Claims.xlsx`	Source Dataset (Frozen)	Shared	4,500 claims x 19 cols (`6.00%` fraud, SHA-256 `952b6e2...45`). Never modified.
`Approch-1/`	Classical ML (`ML`)	J. Ganesh Kumar Reddy (`23BDS024`)	EDA, preprocessing (`29f`), training (`LR`, `RF`, `XGB`, `SVM`, `HGB`), saved models, evaluation JSONs/PNGs.
`Approch-2-DL/`	Deep Learning (`DL`)	M. Jagadeeswar (`23BDS033`)	`MLP`, `ResidualMLP`, `CNN1D`, `TabTransformer`, `Autoencoder` checkpoints (`PyTorch`) + metrics/smoke tests.
`Approach-3-Agentic-AI/`	Agentic & XAI Scaffold	B. Varshith (`23BDS011`)	Port `:8020` FastAPI prototype (`predict.py` deterministic formula, `XAIExplainer` SHA-256 + SHAP-approx helper).
`ml-service/`	Unified Inference Server (`:8000`)	M. Jagadeeswar (`23BDS033`)	Production FastAPI service exposing `/predict/classical/{model}` (5 models) and `/predict/dl/{model}` (5 models).
`backend/node-api/`	Express API Gateway (`:3000`)	M. Jagadeeswar & B. Varshith	TypeScript Express server: proxies ML/DL to `:8000`, serves `/agentic/predict` (`AgenticEnsemble-v1`) & Gemini `/api/chat`.
`frontend/`	Interactive Web App (`:5173`)	M. Jagadeeswar & B. Varshith	React 19 + Vite + TypeScript + Tailwind v4 UI: Home, Classical ML, DL, Agentic Workflow (`6-agent` cards) & Chatbox.
`submission/`	Controlled Research Benchmark & Browser Lab (`:8080`)	J. Ganesh Kumar Reddy (`23BDS024`)	32-run pipeline (`8 models x 4 experiments`), bootstrap CI, exported browser weights (`<4.42e-7` parity), static site.
`docs/`	Project Documentation Suite	Unified Team Deliverable	Literature Survey, Problem Formulation, Innovation & Implementation, Full Documentation.
`deliverables/`	Reports, IEEE Paper & Slides	Unified Team Deliverable	IIIT Dharwad LaTeX Report + PDF, IEEE LaTeX + PDF + DOCX Paper, 20-Slide Presentation Deck + PDF, 4 Standalone Doc PDFs, High-DPI Figures.
`PROJECT_PROGRESS.md`, `DECISIONS.md`	Engineering Contracts & Logs	Unified Team	Canonical 29-feature vs. Track B boundaries, uncalibrated score semantics, and benchmark snapshots.

2. Two Complementary Verification Tracks (`Track A` vs. `Track B`)

1. **Track A — Full-Stack 10-Model Contract (`Approch-1/`, `Approch-2-DL/`, `ml-service/`, `backend/`, `frontend/`):**

- Evaluates all **5 Classical ML** models and all **5 Deep Learning** models on the canonical 29-feature baseline contract (`Train=3,150 [189]`, `Val=675 [41]`, `Test=675 [40]`, seed `42`, including `Cluster`).
- Supervised classification cutoff reported at standard `0.50` (`Approch-2-DL/results/*_metrics.json` and `Approch-1/evaluation/model_results.json`), plus Autoencoder at validation 95th-percentile reconstruction

MSE `0.128588`.

2. **Track B — Leakage-Governed Research Lab (`submission/`):**

- Evaluates all **8 supervised models** (`LR`, `RF`, `XGB`, `SVM`, `HGB`, `MLP`, `ResidualMLP`, `CNN1D`) across **4 protocols** (`baseline 29f`, `intake 28f without Cluster`, `no_amount 27f without Cluster & ClaimAmount`, and `temporal intake >= 2024-03-11`) — `32` **runs /** `128` **val+test metric rows**.
- Optimizes the decision threshold strictly on the validation set (`validation_f1` policy) and exports pure-JSON browser weights verified against Python.

3. Complete Verified Quantitative Results

3.1 Track A — Held-Out Test Benchmarks (29 Features, `N=675`, 40 Frauds)

Model	Family	Test Acc (%)	Prec (%)	Rec (%)	F1 (%)	ROC-AUC (%)	PR-AUC / AP (%)	Confusion (TN/FP/FN/TP)	Cutoff
XGBoost	Classical ML	99.85	97.56	100.00	**98.77**	**100.00**	**100.00**	`634 / 1 / 0 / 40`	`0.50`
RandomForest	Classical ML	99.85	100.00	97.50	98.73	99.996	99.94	`635 / 0 / 1 / 39`	`0.50`
HistGradientBoosting	Classical ML	99.85	100.00	97.50	98.73	98.47	97.65	`635 / 0 / 1 / 39`	`0.50`
TabTransformer	Supervised DL	98.67	81.63	100.00	89.89	**99.94**	**99.11**	`626 / 9 / 0 / 40`	`0.50`
CNN1D	Supervised DL	98.81	83.33	100.00	**90.91**	99.92	98.82	`627 / 8 / 0 / 40`	`0.50`
ResidualMLP	Supervised DL	98.52	80.00	100.00	88.89	99.75	95.61	`625 / 10 / 0 / 40`	`0.50`
MLP	Supervised DL	98.81	84.78	97.50	90.70	99.73	95.38	`628 / 7 / 1 / 39`	`0.50`
SVM (RBF)	Classical ML	97.78	74.51	95.00	83.52	99.55	92.52	`622 / 13 / 2 / 38`	`0.50`
LogisticRegression	Classical ML	96.44	62.50	100.00	76.92	99.50	91.86	`611 / 24 / 0 / 40`	`0.50`
Autoencoder *(Neg. Result)*	Unsupervised DL	90.37	3.70	2.50	**2.99**	46.02	**5.23**	`609 / 26 / 39 / 1`	MSE `0.1286`

3.2 Track B — Four-Experiment Sensitivity & Chronological Ablation (`validation_f1` Policy)

Model	Baseline Test AP / F1 (`29f`)	Intake Test AP / F1 (`28f`, No Cluster)	No-Amount Test AP / F1 (`27f`)	Temporal Intake Test AP / F1 (`>=2024-03-11`)	Intake Errors (FP / FN) & tau*
RandomForest *(Selected Default)*	`0.9994 / 0.9750`	`1.0000 / 0.9873`	`0.2066 / 0.3636`	`1.0000 / 1.0000`	`FP=0, FN=1`, tau*=0.4167
XGBoost	`1.0000 / 0.9877`	`1.0000 / 0.9877`	`0.2128 / 0.3524`	`1.0000 / 1.0000`	`FP=1, FN=0`, tau*=0.2121
GradientBoosting (`HGB`)	`0.9765 / 0.9873`	`0.9765 / 0.9873`	`0.2404 / 0.3636`	`1.0000 / 1.0000`	`FP=0, FN=1`, tau*=1.0000
CNN1D	`0.9882 / 0.9268`	`0.9667 / 0.8767`	`0.2287 / 0.2788`	`0.9172 / 0.8108`	`FP=1, FN=8`, tau*=0.9691
ResidualMLP	`0.9567 / 0.9286`	`0.8765 / 0.8315`	`0.2246 / 0.3438`	`0.7979 / 0.7229`	`FP=12, FN=3`, tau*=0.2751
MLP	`0.9538 / 0.9136`	`0.8866 / 0.8222`	`0.2303 / 0.3077`	`0.8581 / 0.6957`	`FP=13, FN=3`, tau*=0.8359
LogisticRegression	`0.9186 / 0.7835`	`0.8525 / 0.7327`	`0.2318 / 0.3191`	`0.8799 / 0.7397`	`FP=24, FN=3`, tau*=0.7618
SVM (RBF)	`0.9252 / 0.8974`	`0.8181 / 0.7711`	`0.1869 / 0.3041`	`0.8396 / 0.7619`	`FP=11, FN=8`, tau*=0.5859

4. Setup & Developer Runbook

4.1 Prerequisites

- Python `>= 3.10` (`scikit-learn`, `xgboost`, `torch`, `pandas`, `openpyxl`, `fastapi`, `uvicorn`, `pydantic`)
- Node.js `>= 18` & `npm`
- Optional: `GEMINI_API_KEY` environment variable in `backend/node-api/.env` for live `/api/chat` responses

4.2 Running Path A — Three-Tier Enterprise Application

Start each tier in a dedicated terminal from repository root:

```
# Terminal 1: Unified FastAPI Inference Service (:8000)
cd ml-service
pip install -r requirements.txt
python -m uvicorn app.main:app --host 127.0.0.1 --port 8000

# Terminal 2: Node.js + Express Gateway & Agentic/Gemini Server (:3000)
cd backend/node-api
npm install
# Optional: set GEMINI_API_KEY in .env
npm run dev

# Terminal 3: React + Vite Dashboard (:5173)
cd frontend
npm install
npm run dev
```

4.3 Running Path B — Reproducible Benchmark & Static Browser Lab (`submission/`)

```
# Re-run full 32-run benchmark + browser weight export + verification
python submission/scripts/run_pipeline.py
python submission/scripts/make_figures.py
python submission/scripts/export_browser.py
python submission/scripts/verify.py

# Serve Claim Fraud Lab locally on :8080
python submission/scripts/serve.py
```

5. API & Schema Reference

5.1 FastAPI Model Inference (`ml-service` on `http://127.0.0.1:8000`)

- `GET /health`: Service readiness + loaded Classical & Deep Learning model inventory.
- `POST /predict/classical/{model_name}` where `model_name in {logistic_regression, random_forest, xgboost, svm, gradient_boosting}` (or UI alias mapping via Express).
- `POST /predict/dl/{model_name}` where `model_name in {mlp, residual_mlp, cnn1d, tabtransformer, autoencoder}`.

Standard Request Body (`ClaimFeaturesRequest`):

```
{
  "ClaimAmount": 8500.0,
  "PatientAge": 45,
  "PatientIncome": 48000.0,
  "ClaimDate": "2024-03-15",
  "PatientGender": "M",
  "ProviderSpecialty": "Cardiology",
  "ClaimStatus": "Pending",
  "PatientMaritalStatus": "Married",
  "PatientEmploymentStatus": "Employed",
  "ClaimType": "Inpatient",
  "ClaimSubmissionMethod": "Online",
  "Cluster": 1
}
```

5.2 Node.js Express Gateway (`backend/node-api` on `http://127.0.0.1:3000`)

- `GET /health`: Gateway health & downstream connectivity.
- `POST /api/ml/predict`: Proxies payload cleanly to `:8000` classical or DL endpoint.
- `POST /agentic/predict`: Executes `AgenticEnsemble-v1` (4 trace events: `ingestion`, `validation`, `risk-scorer`, `xai-explainer`) over 9 intake attributes.
- `POST /api/chat`: Calls Google Gemini (`gemini-2.0-flash`) with domain-grounded claim context (`backend/node-api/src/routes/chat.ts`).

6. Documented Limitations & Stale Artifact Discrepancies (Honest Engineering Notes)

1. `Cluster` **Leakage in Track A vs. Track B:** Track A keeps `Cluster` (`29f`) strictly to match historical checkpoint shapes and API schemas; real-time claim intake before post-hoc clustering MUST use Track B's `Intake (28f)` pipeline or drop `Cluster`.
2. `ClaimAmount` **Single-Feature Sensitivity:** Dropping `ClaimAmount` in `no_amount` lowers every classifier from `~1.0000` AP down to `0.1869–0.2404` AP because dataset positive labels never occur below `$7,465.07`. Future synthetic/real clinical datasets must inject low-dollar anomalies (e.g., high-frequency upcoding) to avoid amount shortcuts.
3. **Uncalibrated Output Scores:** Both neural logits passed through `sigmoid` under `pos_weight=15.67` and tree/Platt outputs represent ranking propensities rather than well-calibrated posterior probabilities (`CalibratedClassifierCV` is recommended before monetary expected-loss multiplication).
4. **Approach 3 Runtime vs. Research UI Separation:** `backend/node-api/src/agentic/predict.ts` currently returns a fast deterministic heuristic formula rather than calling `:8000` FastAPI ML/DL models or running LangGraph/RAG/OCR pipelines; `Approach-3-Agentic-AI/app/services/xai_service.py` provides an offline XAI/SHA-256 scaffold awaiting full service unification.
5. **Submodule README Updates:** Legacy developer notes inside `Approch-2-DL/README.md` and `backend/node-api/README.md` reflect earlier sprint snapshots; readers should rely on this unified documentation suite (`docs/`), `PROJECT_PROGRESS.md`, `DECISIONS.md`, and live source code.

Supporting Visual Architecture & Quantitative Figures

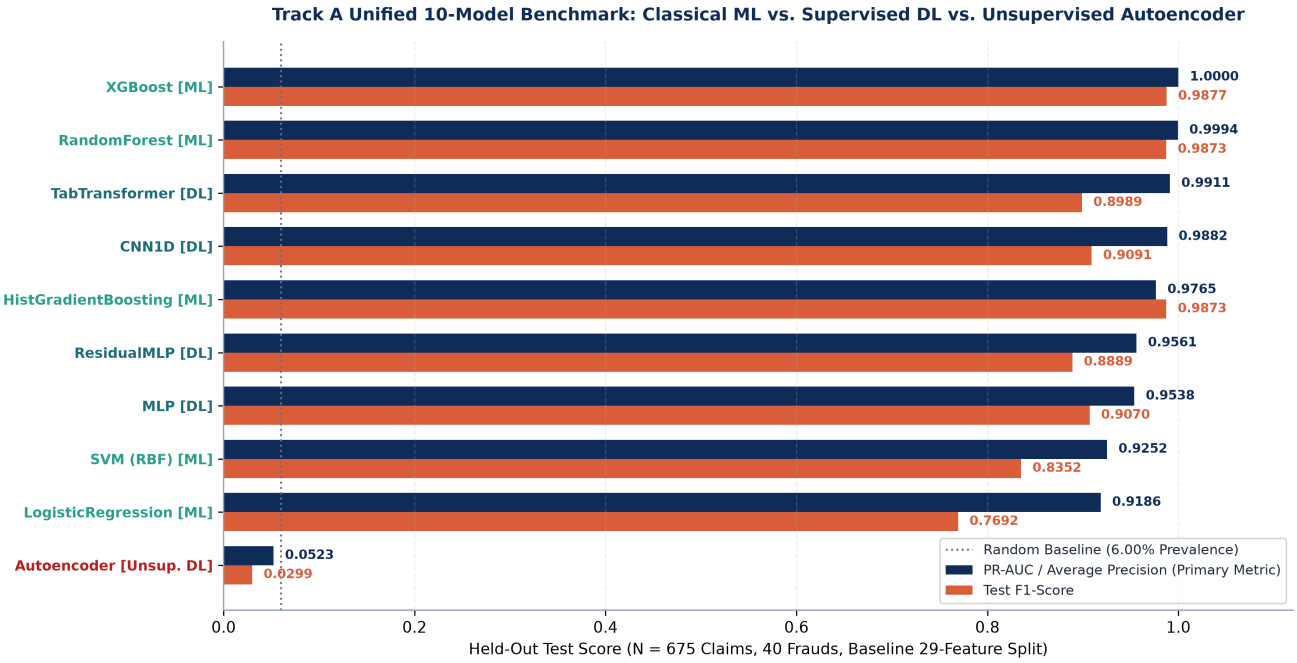


Figure 1: Track A 10-model PR-AUC vs. F1 benchmark (29 features).

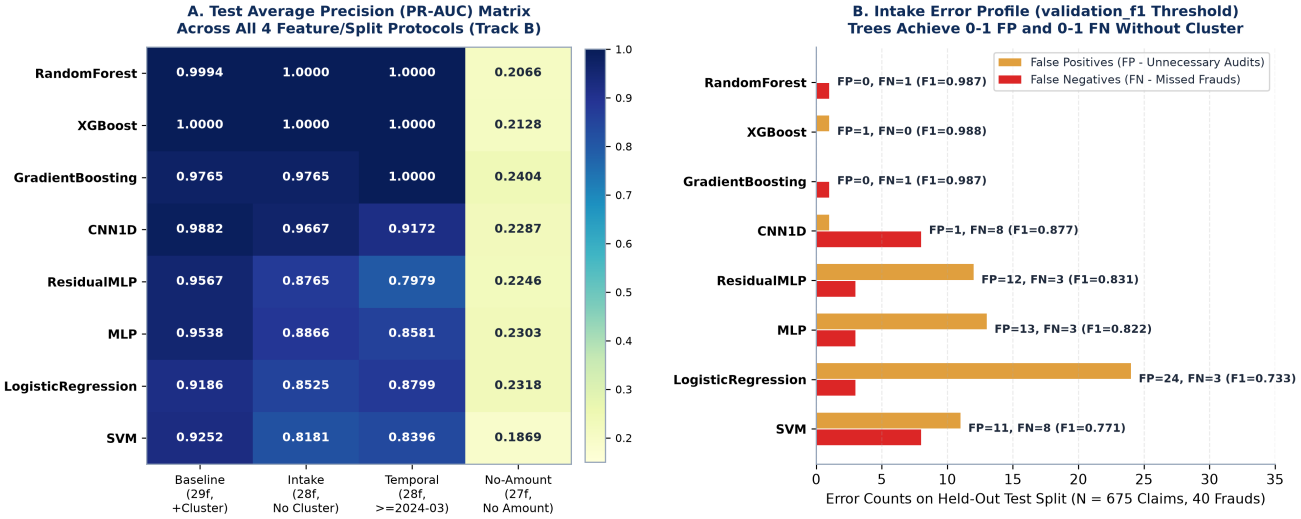


Figure 2: Track B 4-experiment PR-AUC ablation & Intake FP/FN error counts.

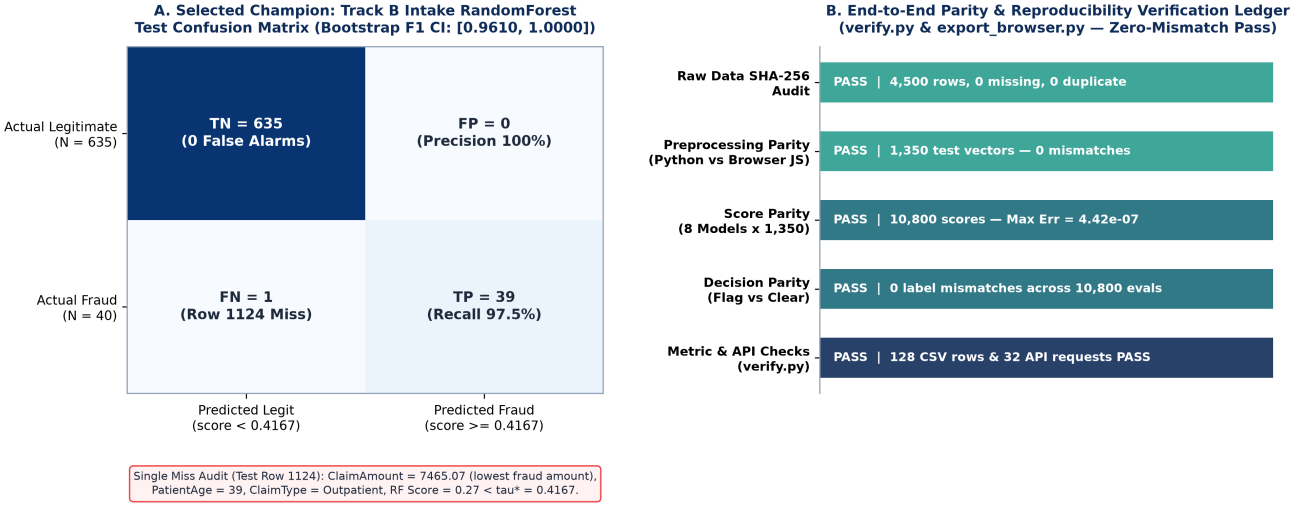


Figure 3: Champion Intake RandomForest error audit & Python-to-Browser parity verification.